

3 (2)

Histogram and Frequency Polygon

Grouped Frequency Distribution(Contd. . .)

83	29	40	48	55	59	61	66	73
75	61	42	85	59	46	32	88	79
61	54	65	63	26	85	38	77	49
68	75	60	41	62	90	38	54	61
68	71	58	64	77	50	56	63	73
33	46	51	71	54	64	65	63	75



Grouped Frequency Distribution(Contd. . .

<u>83</u>	<u>75</u>	<u>65</u>	<u>85</u>	<u>77</u>	<u>90</u>	<u>65</u>	<u>88</u>	<u>79</u>
83	29	40	48	55	59	61	66	73
75	61	42	85	59	46	32	88	79
61	54	65	63	<u>26</u>	85	38	77	49
68	75	60	41	62	<u>90</u>	38	54	61
68	71	58	64	77	50	56	63	73
<u>33</u>	<u>46</u>	<u>51</u>	<u>71</u>	<u>54</u>	<u>64</u>	<u>65</u>	<u>63</u>	<u>75</u>
33	29	42	41	26	46	32	54	61

Grouped Frequency Distribution(Contd. . . .)

Since we are asked to construct a GFD
u classes and Since $x_{max} = 90$ and
 $x_{min} =$ follows that

$$\begin{aligned}\text{Class width} &= \frac{x_{max} - x_{min}}{\text{Number of classes}} = \frac{90 - 26}{6} \\ &= \frac{64}{6} = 10.66666667 \approx 11\end{aligned}$$

Grouped Frequency Distribution(Contd. . . .)

Now we are going to construct a GFD with 6 classes starting from the smallest value.

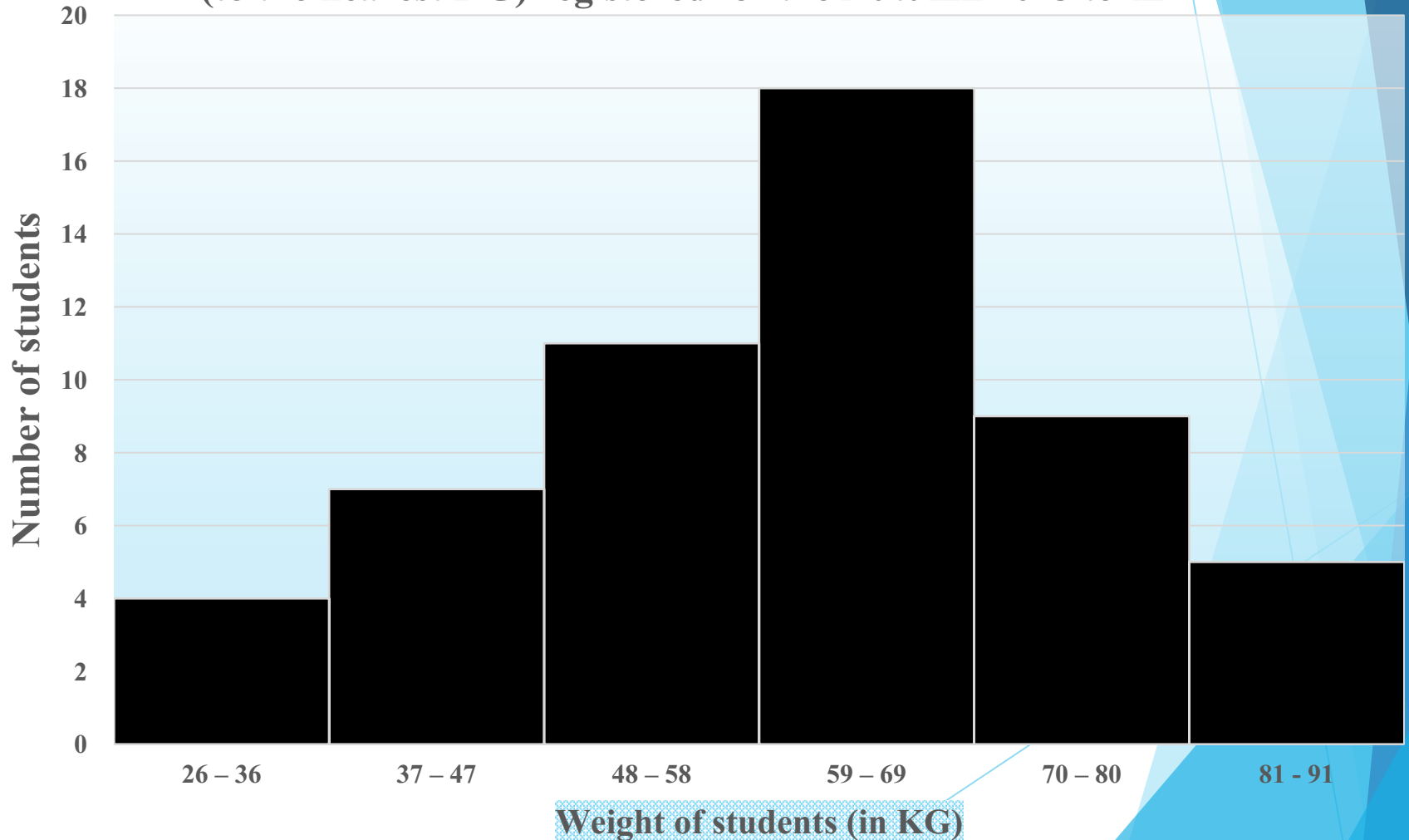
	Class Limit	Frequency	Class Boundary	Class Mark
1 st Class	26 - 36	4	25.5 - 36.5	31
2 nd Class	37 - 47	7	36.5 - 47.5	42
3 rd Class	48 - 58	10	47.5 - 58.5	53
4 th Class	59 - 69	18	58.5 - 69.5	64
5 th Class	70 - 80	10	69.5 - 80.5	75
6 th Class	81 - 91	5	80.5 - 91.5	86

Histogram and Frequency Polygon

- ▶ A histogram is a graphical representation that organizes a group of data points into user-specified ranges. It is similar in appearance to a bar graph. The histogram condenses a data series into an easily interpreted visual by taking many data points and grouping them into logical ranges or intervals.

Histogram and Frequency Polygon (Contd. . . .)

Histogram presenting the weight of a sample of HiLCoE students
(to the nearest KG) registered for the Autumn 2025 term

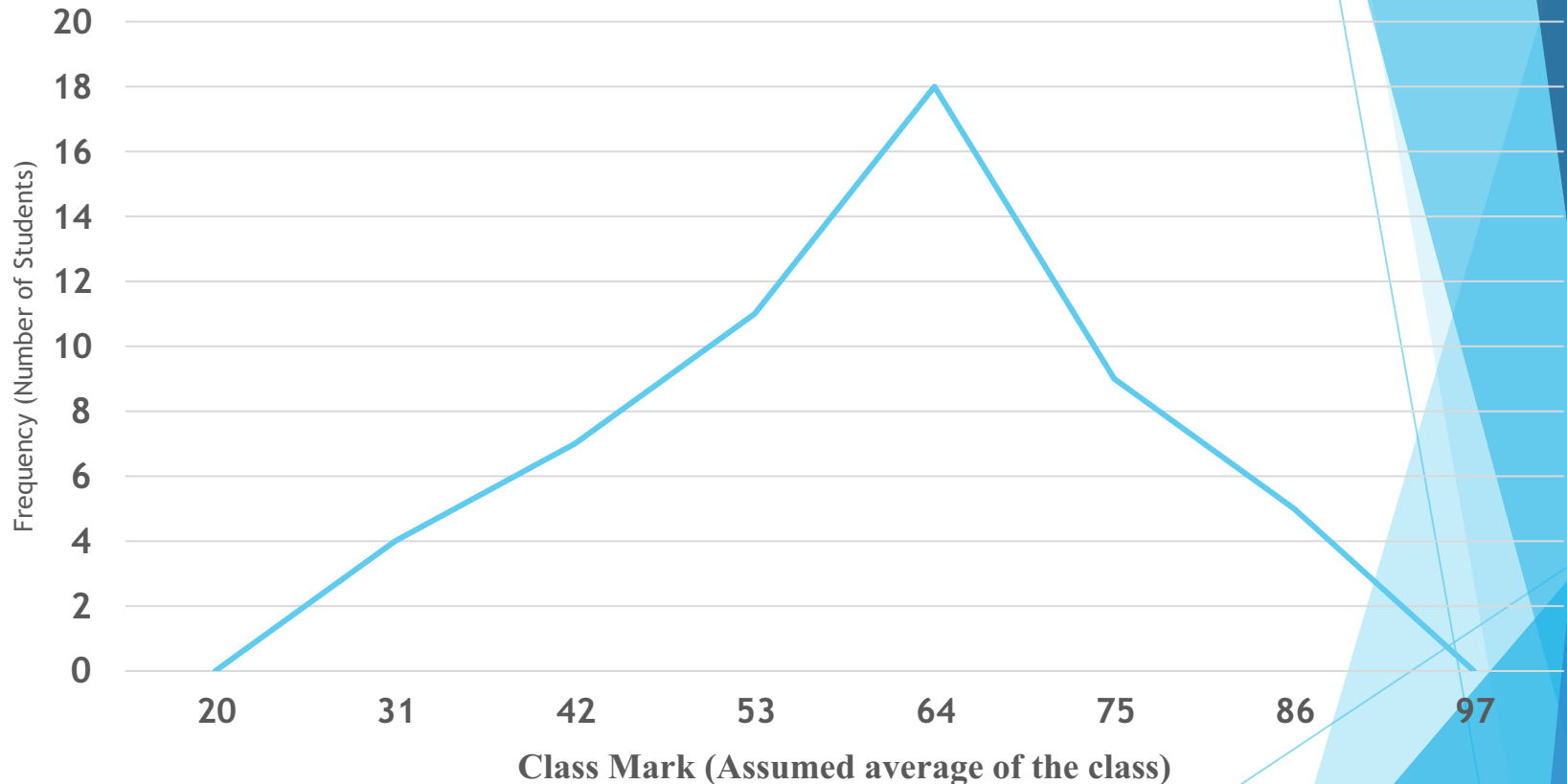


Histogram and Frequency Polygon (Contd. . . .)

Another type of graph that can be drawn to represent the same set of data as a histogram represents is a frequency polygon. A frequency polygon is a graph constructed by using lines to join the midpoints of each interval which we call *class mark*. The heights of the points represent the frequencies.

Histogram and Frequency Polygon (Contd. . . .)

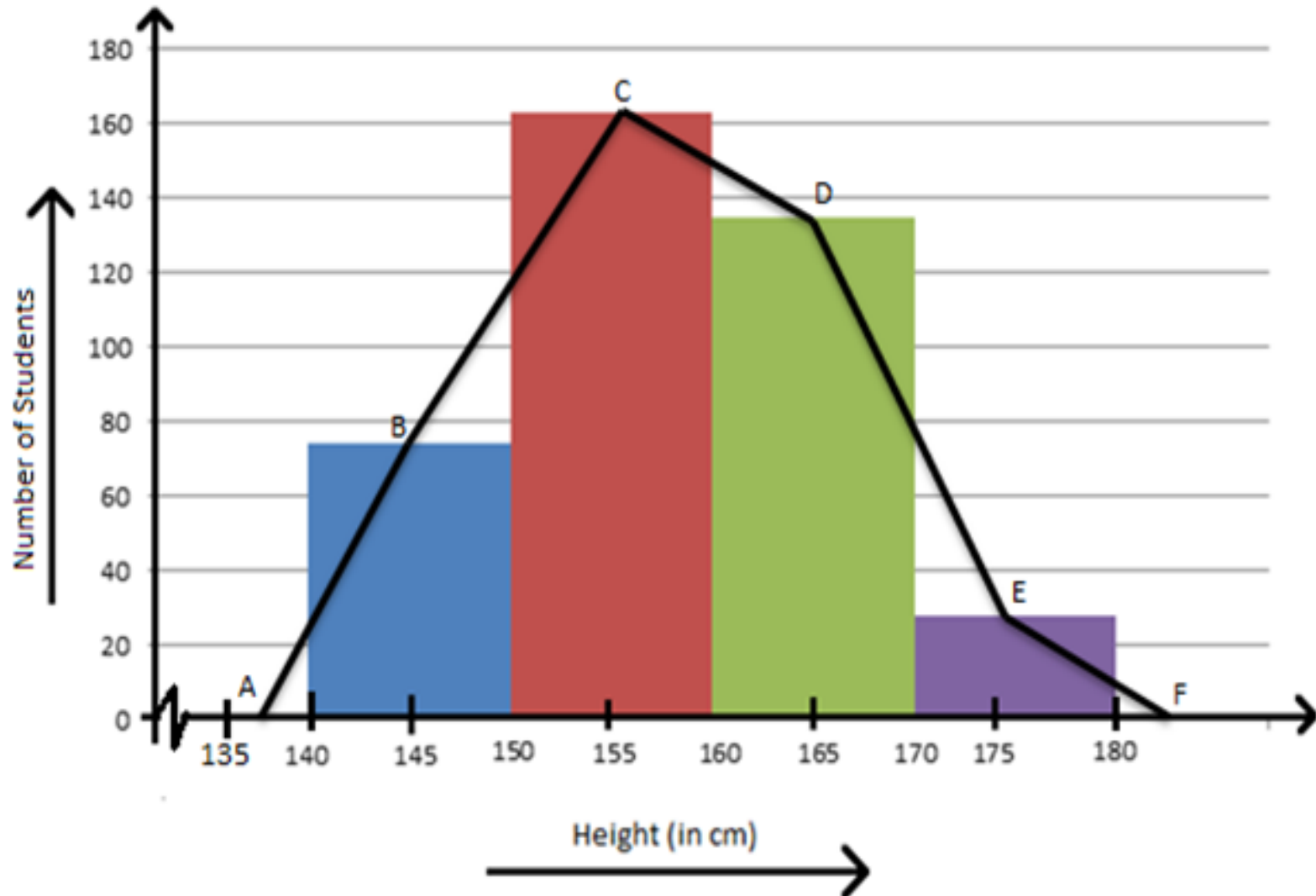
Frequency Polygon presenting the weight of the sample of 54 students registered for the Autumn 2024 term



Histogram and Frequency Polygon (Contd. . . .)

- ▶ A **frequency polygon** can be created from the histogram or by joining the midpoints of the upper edges of the rectangles in a histogram whose class intervals are of uniform length.

Histogram and Frequency Polygon (Contd. . . .)



$$3(3)^{-2}$$

**Relative Frequency
Distribution and
Cumulative Frequency
Distribution**

Relative Frequency Distribution

The relative frequency of a class is defined by:

$$\text{RF of a class} = \frac{\text{Frequency of a class}}{\text{Sum of all frequencies}}$$

And the relative frequency percentage is defined by:

$$\text{RFP} = \text{RF} \times 100\%$$

Relative Frequency Distribution (Contd. . . .)

Now consider the GFD given in example 2.

Class Limit	Frequency	Relative Frequency	RFP
26 - 36	4	$\frac{4}{54} = 0.074$	$0.074 \times 100\% = 7.4 \%$
37 - 47	7	$\frac{7}{54} = 0.13$	$0.13 \times 100\% = 13 \%$
48 - 58	10	$\frac{10}{54} = 0.185$	$0.185 \times 100\% = 18.5 \%$
59 - 69	18	$\frac{18}{54} = 0.333$	$0.333 \times 100\% = 33.3 \%$
70 - 80	10	$\frac{10}{54} = 0.185$	$0.185 \times 100\% = 18.5 \%$
81 - 91	5	$\frac{5}{54} = 0.093$	$0.093 \times 100\% = 9.3 \%$
Total	54	1	100 %

Cumulative Frequency Distribution and its graph

Cumulative Frequency Distribution

The total frequency of all classes less than (or more than or equal to) the upper-class boundary of a given class is called the cumulative frequency of that class. A table showing the cumulative frequencies is called a *cumulative frequency distribution*. It means we have two types of cumulative frequency distributions:

Less than type CFD *and* More than type CFD

Cumulative Frequency Distribution (Contd. . . .)

► *Less than cumulative frequency distribution:*

It is obtained by adding successively the frequencies of all the previous classes including the class against which it is written. The cumulate is started from the lowest to the highest size.

Cumulative Frequency Distribution (Contd. . . .)

? More than cumulative frequency distribution:

It is obtained by finding the cumulate total of frequencies starting from the highest to the lowest class. In both cases we shall consider the class boundaries for continuous data.

Now consider the GFD given in example 2 and construct a CFD of both types.

Cumulative Frequency Distribution (Contd. . . .)

Class Boundary	Frequency	CB	CF < type
[25.5 - 36.5)	4	Less than 36.5	4
[36.5 - 47.5)	7	Less than 47.5	11
47.5 - 58.5	10	Less than 58.5	21
58.5 - 69.5	18	Less than 69.5	39
69.5 - 80.5	10	Less than 80.5	49
80.5 - 91.5	5	Less than 91.5	54

Class Boundary	Frequency	CB	CF > type
[25.5 - 36.5)	4	25.5 or more	54
[36.5 - 47.5)	7	36.5 or more	50
47.5 - 58.5	10	47.5 or more	43
58.5 - 69.5	18	58.5 or more	33
69.5 - 80.5	10	69.5 or more	15
80.5 - 91.5	5	80.5 or more	5

Cumulative Frequency Distribution (Contd. . . .)

Less than type CFD

Class B0undaries	CF (Less than type)
Less than 25.5	0
Less than 36.5	4
Less than 47.5	11
Less than 58.5	22
Less than 69.5	39
Less than 80.5	49
Less than 91.5	54

Cumulative Frequency Distribution (Contd. . . .)

More than type CFD

Class Boundaries	CF (More than type)
25.5 or more	54
36.5 or more	50
47.5 or more	43
58.5 or more	32
69.5 or more	14
80.5 or more	5
91.5 or more	0

Cumulative Percentage Frequency Distribution

A cumulative percentage frequency distribution is a cumulative frequency distribution in which the cumulative frequencies are expressed in *percentage*.

Consider the CFD of both types discussed above, construct cumulative percentage frequency distributions.

Cumulative Percentage Frequency Distribution (Contd. . . .)

Class Boundaries	Cumulative Frequency	Cumulative Percentage
Less than 25.5	0	$\frac{0}{54} \times 100\% = 0\%$
Less than 36.5	4	$\frac{4}{54} \times 100\% = 7.41\%$
Less than 47.5	11	$\frac{11}{54} \times 100\% = 20.37\%$
Less than 58.5	21	$\frac{22}{54} \times 100\% = 40.74\%$
Less than 69.5	39	$\frac{39}{54} \times 100\% = 72.2\%$
Less than 80.5	49	$\frac{49}{54} \times 100\% = 90.74\%$
Less than 91.5	54	$\frac{54}{54} \times 100\% = 100\%$

Cumulative Percentage Frequency Distribution (Contd. . . .)

Class Boundaries	Cumulative Frequency	Cumulative Percentage
25.5 or more	54	$\frac{54}{54} \times 100\% = 100\%$
36.5 or more	50	$\frac{50}{54} \times 100\% = 92.59\%$
47.5 or more	43	$\frac{43}{54} \times 100\% = 79.63\%$
58.5 or more	33	$\frac{32}{54} \times 100\% = 59.26\%$
69.5 or more	13	$\frac{14}{54} \times 100\% = 25.93\%$
80.5 or more	5	$\frac{5}{54} \times 100\% = 9.26\%$
91.5 or more	0	$\frac{0}{54} \times 100\% = 0\%$

The graph of a CFD (Ogive)

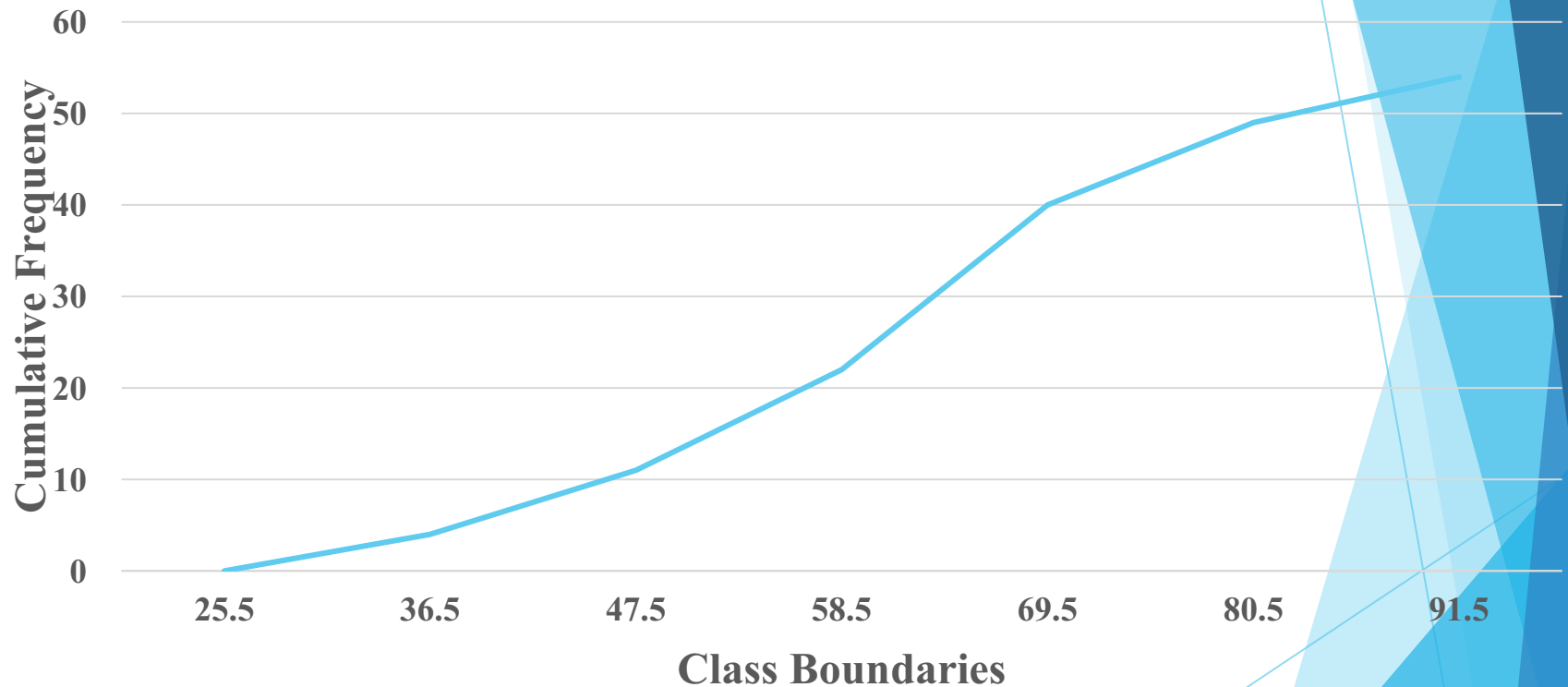
The graph of a CFD is called an Ogive which is read as “OJAIVE.”

As we have two types of CFD, we will have two types of Ogives:

- ? Less than type Ogive
- ? More than type Ogive

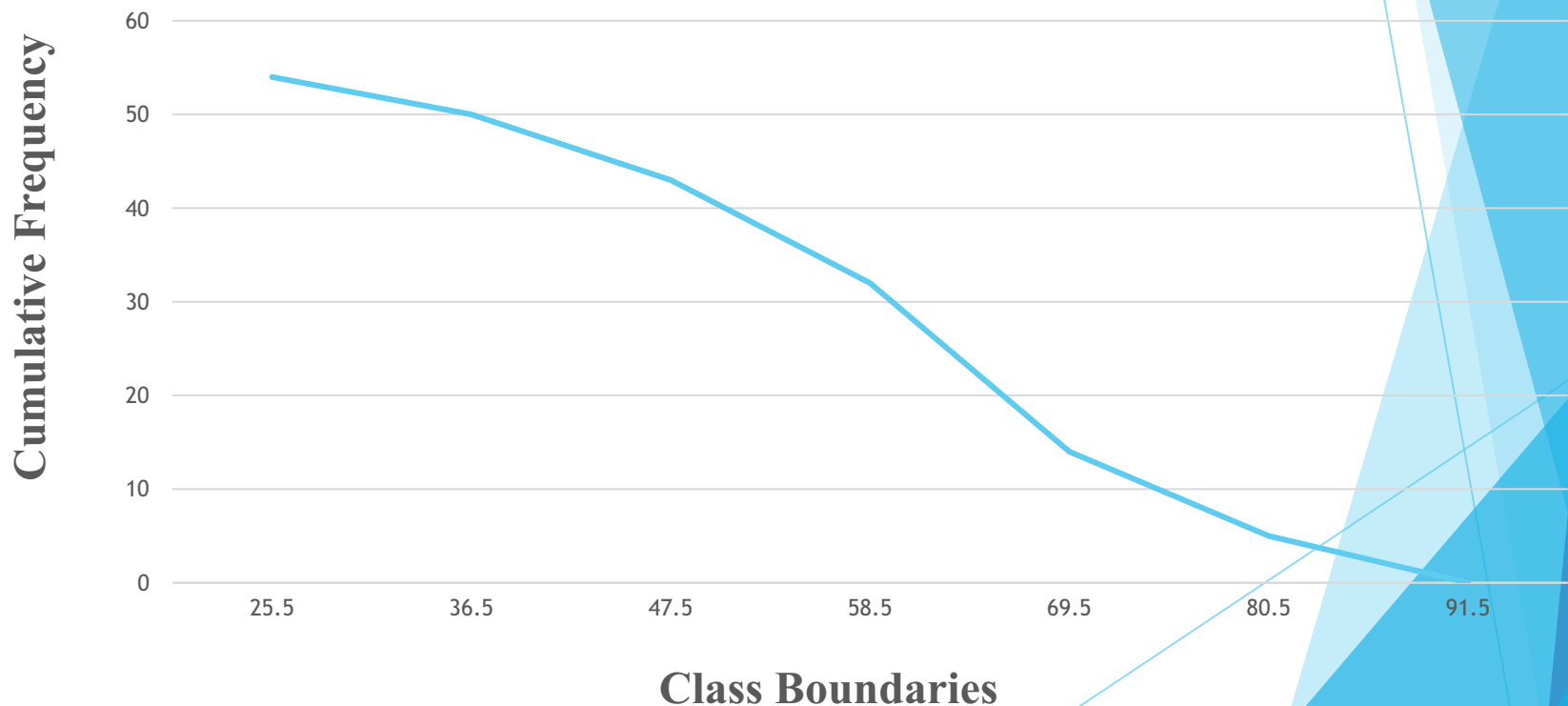
The graph of a CFD (Ogive) (Contd. . . .)

Ogive presenting the weight (to the nearest Kg) of a sample of
HiLCoE students registered for the Spring 2025 Program



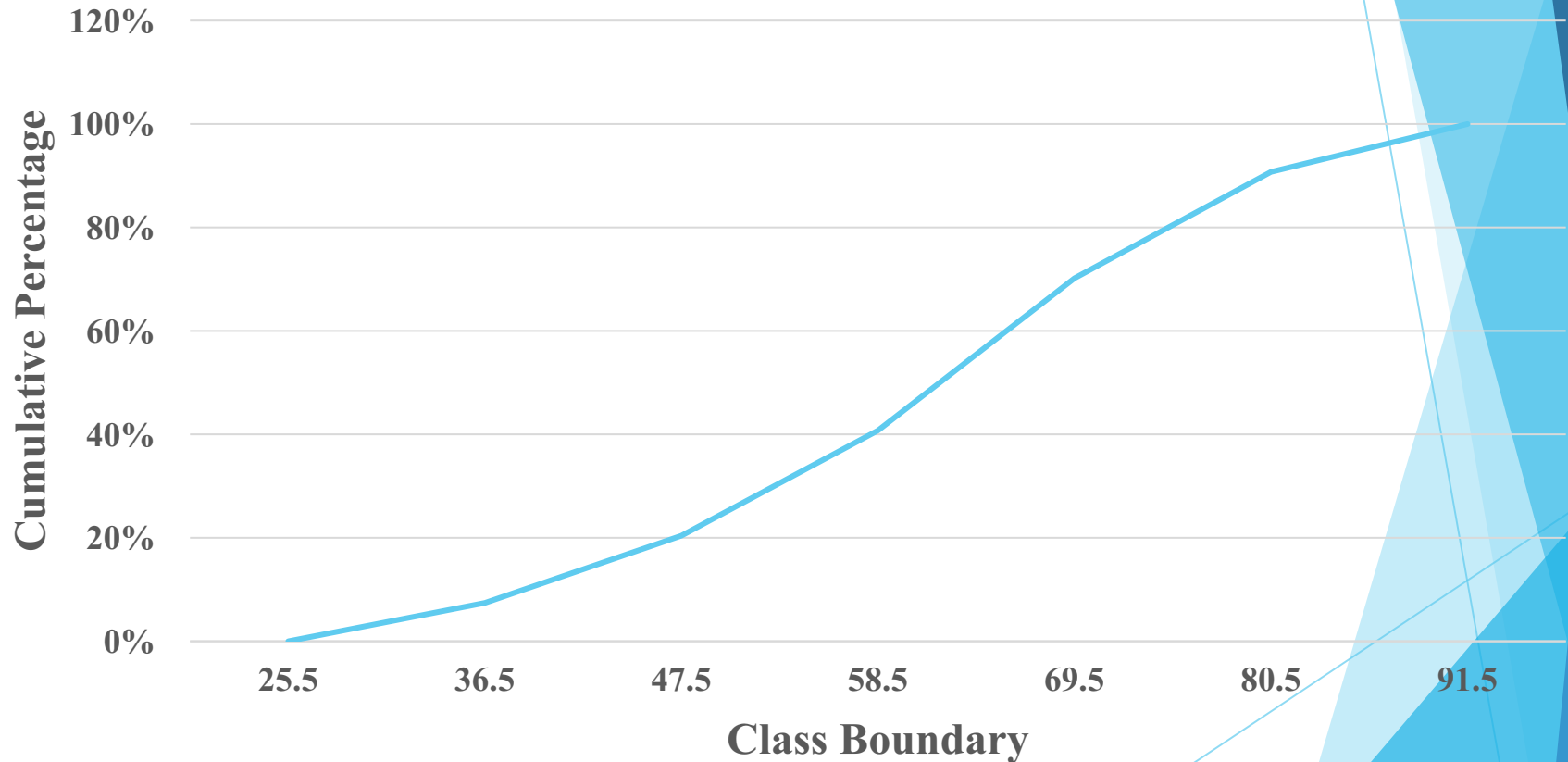
The graph of a CFD (Ogive) (Contd. . . .)

Ogive presenting the weight (to the nearest Kg) of a sample of
HiLCoE students registered for the Spring 2025 Program



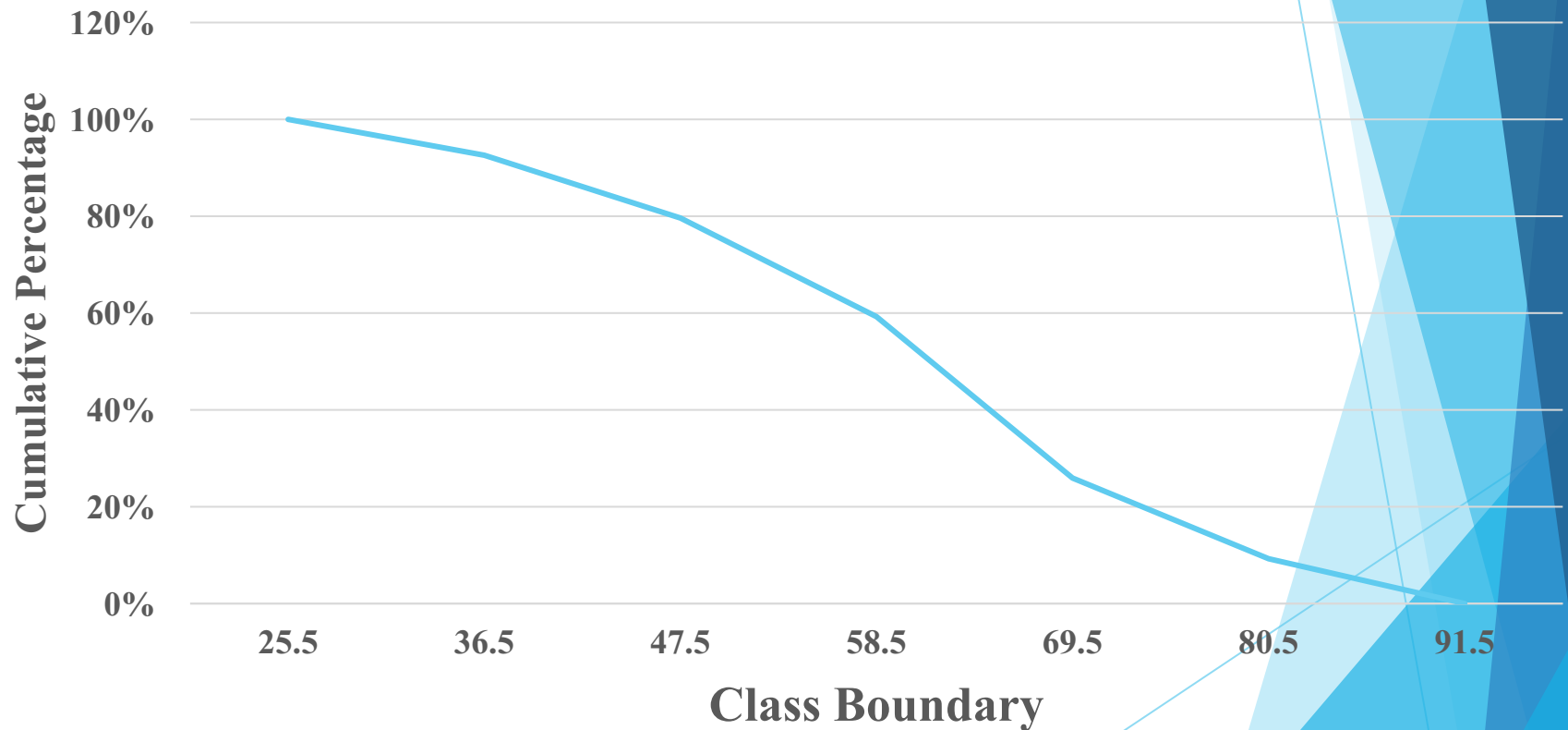
The graph of a CPFD (Percentage Ogive)

Percentage Ogive presenting the weight (to the nearest Kg) of a sample of
HiLCoE students registered for the Spring 2025 Program



The graph of a CPFD (Percentage Ogive)

Percentage Ogive presenting the weight (to the nearest Kg) of a sample of
HiLCoE students registered for the Spring 2025 Program



Any Questions



Any Questions

